Exact censuses against five ECDLP hardness assumptions

Date 2026-11-04. Study `research/ecdlp_weakness_census_20261104/`; Lean companions in `research/ecdlp_weakness_lean_20261104/`. Every number is an unverified assertion under the repository's proof-carrying-claims rule (no signed validator report) [pcc:unverified]. Each census is an exact computation in Rust (no dependencies, deterministic seed, in-code self-tests); `plots.py` only lays out recorded values.

What was asked

Attack ten theorems and assumptions that hold up ECDLP hardness, and run the r_2-versus-k rank census that the earlier triage left as the step toward refuting Petit–Quisquater Assumption 2. The owner's aim is to find out whether a subexponential algorithm is possible. Nothing here shows that one exists; the results remove or sharpen individual supports.

The ten attacks

#	target	instrument	outcome
1	`HEUR-GB-PQ-ASSUMPTION-2` (m = 2)	`pq2_rank` census + Lean criterion	literal reading fails on the 111 instances with $n \le 44$; iterated degree > 3 for n = 15 and 17–44, exactly 4 on 33 fresh instances at n = 17–27; msolve F4 degree 4 at n = 17–34
2	degree-2 criterion generalised	Lean `SubspaceCriterion`	$d_{\mathrm{lin}} \le \mathrm{sd}$ for any Macaulay degree; kernel-checked
3	`HEUR-DECOMPOSITION-PROBABILITY` (prime field)	`pf_yield` census	zero class = Poisson once unordered; variance 1.45× Poisson for negation-closed bases
4	mean of the yield heuristic	Lean `MeanCount`	mean $\lvert F \overset{\lvert m}{\cap} G\rvert$ is a theorem
5	`GAP-PF-TRACE-PULLBACK-FACTOR-BASE`	`trace_pullback` census	no dim-1 subspace of $\mathbb{F}_{p^2}$ has a trace image below $0.32 \cdot \#E(\mathbb{F}_p)$
6	`HEUR-BK-GG-DECOMP-PROPORTION` (1/n!)	`ext_decomp` census	0.70–0.99 of $\frac{1}{n!}$ at n = 2, 3; premise fails for subfield curves
7	overbroad Shoup reading, "BSGS optimal"	Lean `TransparentEncoding`	refuted: identity encoding gives the log with zero group operations
8	`HEUR-BK-GG-FACTOR-BASE-SIZE` ($\lvert F\rvert \approx \lvert V\rvert$)	`pq2_rank` census	supported: $\lvert F' \overset{\lvert}{\cap} V\rvert$ in 0.84–1.31 over 99 instances, 0.997–1.004 for $n \ge 30$
9	bridge $k = N - \mathrm{affdim}(Z)$, case $\lvert Z\rvert = 1$	Lean `BooleanNullstellensatz`	unique zero puts every $X_i - z_i$ in the ideal; kernel-checked
10	`HEUR-PF-PKM-RANK-DELTA` ($\Delta = 10$)	`rank_deficit` census	$\Delta = 10$ gives full rank in all samples on 2 of 10 curves; deficits 1–3 persist

pq2_rank

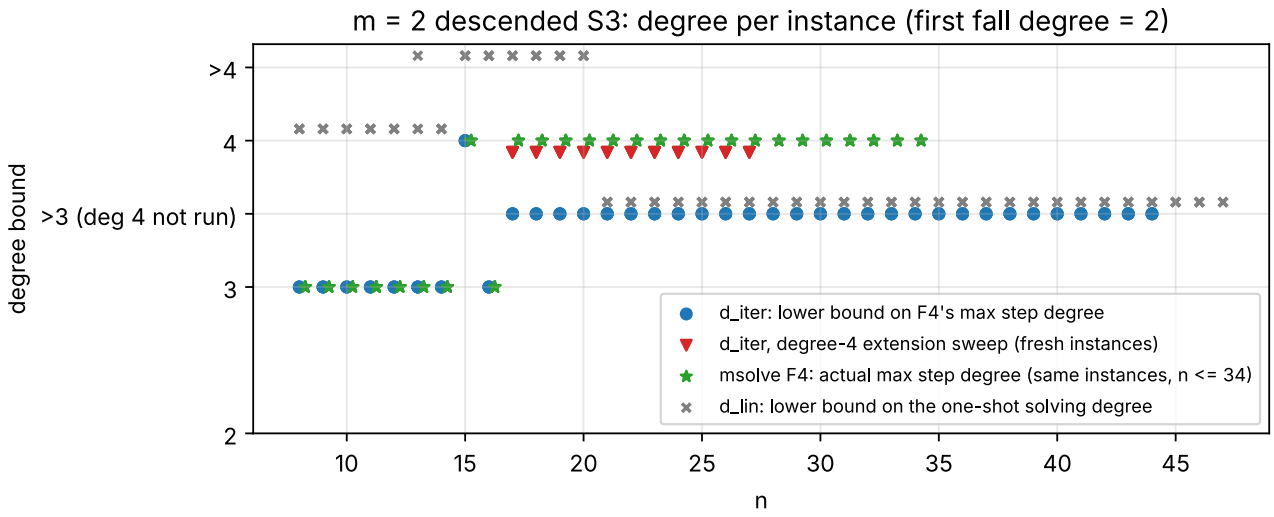


Figure 1: Degree lower bounds per instance. d_iter: least degree whose iterated closure reaches all k linear consequences. d_lin: the same for the one-shot Macaulay matrix.

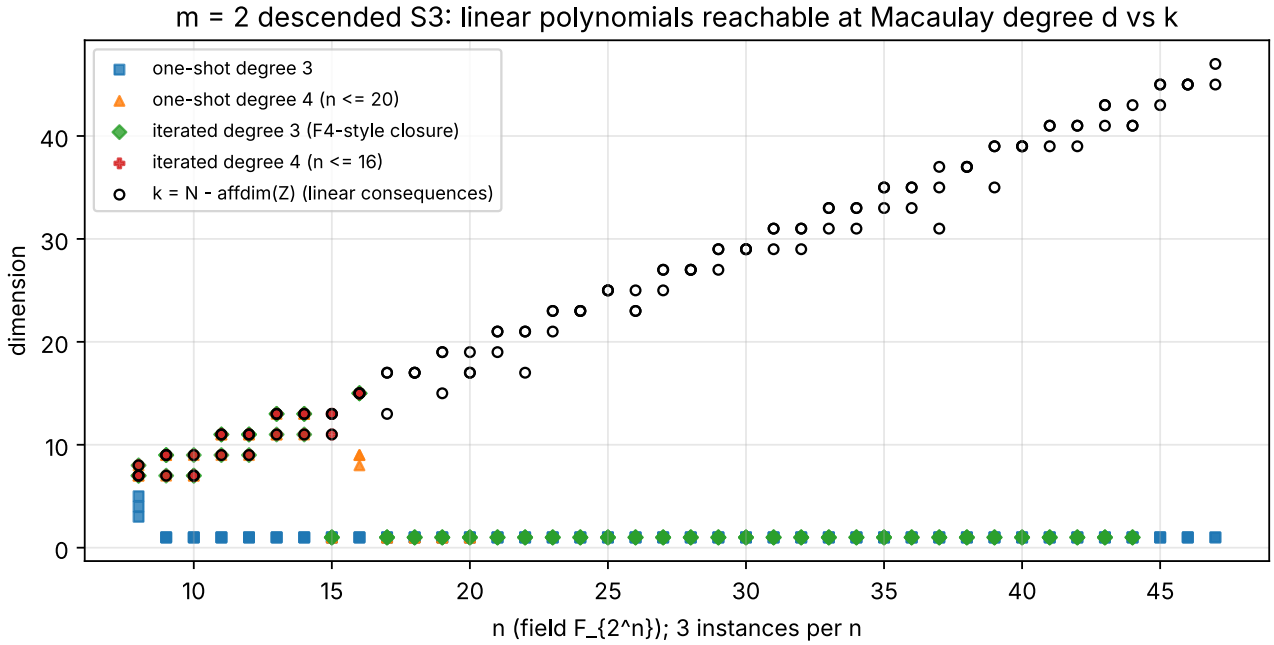


Figure 2: Dimension of the degree-≤1 part of each row space against k.

119 instances, n = 8 to 47, three per n; the sweep stopped during n = 47. The iterated degree-3 closure reaches k for every n ≤ 14 and at n = 16, and reaches only 1 at n = 15 and on all 84 instances from n = 17 to 44. Fresh degree-4 sweeps at n = 17 to 27 reach k on all 33 instances, and msolve's F4 reaches maximal step degree exactly 4 on all 54 main-sweep instances from n = 17 to 34. So the first fall degree 2 is not the solver degree on any instance with $n \leq 44$, and the value 3 of the Kousidis–Wiemers reading is not reached from n = 17 to 44. An independent naive saturation in Python agrees with the Rust closure on five instances, and a same-seed rerun reproduces the log. Lean (`F4Containment`) proves that an F4 run with maximal step degree D stays inside the degree-D closure, so the census values bound F4 from below; $d_{\mathrm{lin}} \leq \mathrm{sd}$ is kernel-checked. The degree 4 matches Kousidis–Wiemers' Magma value for $n \leq 36$; they report 5 from $n = 37$, so a window ending at $n = 34$ does not decide whether the degree stays bounded. m = 2 only. [pcc:unverified]

On fresh instances beyond the census window, msolve gives maximal step degree 4 at $n = 36$. At $n = 37$, run on isolab-eks, it starts a degree-5 F4 step (a 1,016,044 × 610,665 matrix, 53 GB) after four degree-4 steps, reproducing with a second implementation the step to 5 that Kousidis–Wiemers report

for Magma. Kubernetes stopped the run at its deadline about three hours into that step, so the final basis check is unavailable; the corpus records the result as an obstruction to Semaev's $d_{\mathrm{F4}} \leq 4$ (`MEAS-PQ2-MSOLVE-N37-STEP5`). A rerun to completion was not started because the account's monthly budget is exhausted. [pcc:unverified]

pf_yield

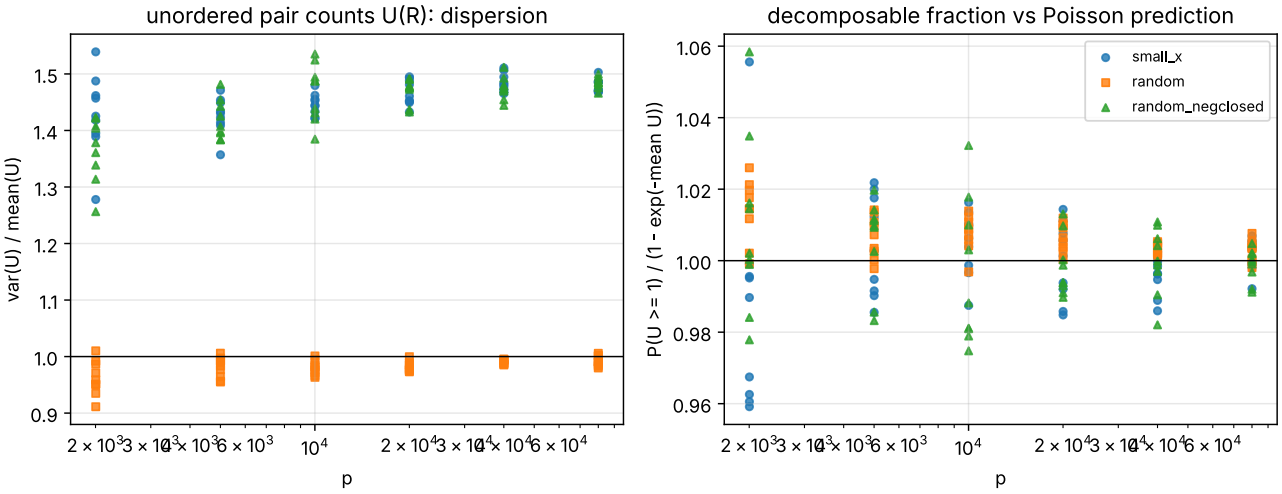


Figure 3: Unordered pair counts U(R): dispersion and zero class, three factor-base shapes, p from 2003 to 80021.

Eighteen prime-order curves, $\mu = \frac{B^2}{r}$ near 0.5, 1, 2, every target enumerated. P(U ≥ 1) matches $1 - \exp(-\mathrm{mean}\ U)$ within about 1% on average for all three base shapes; var/mean of U is about 1.45 for both negation-closed bases and 0.97–0.99 for the random base not closed under negation. [pcc:unverified]

rank_deficit

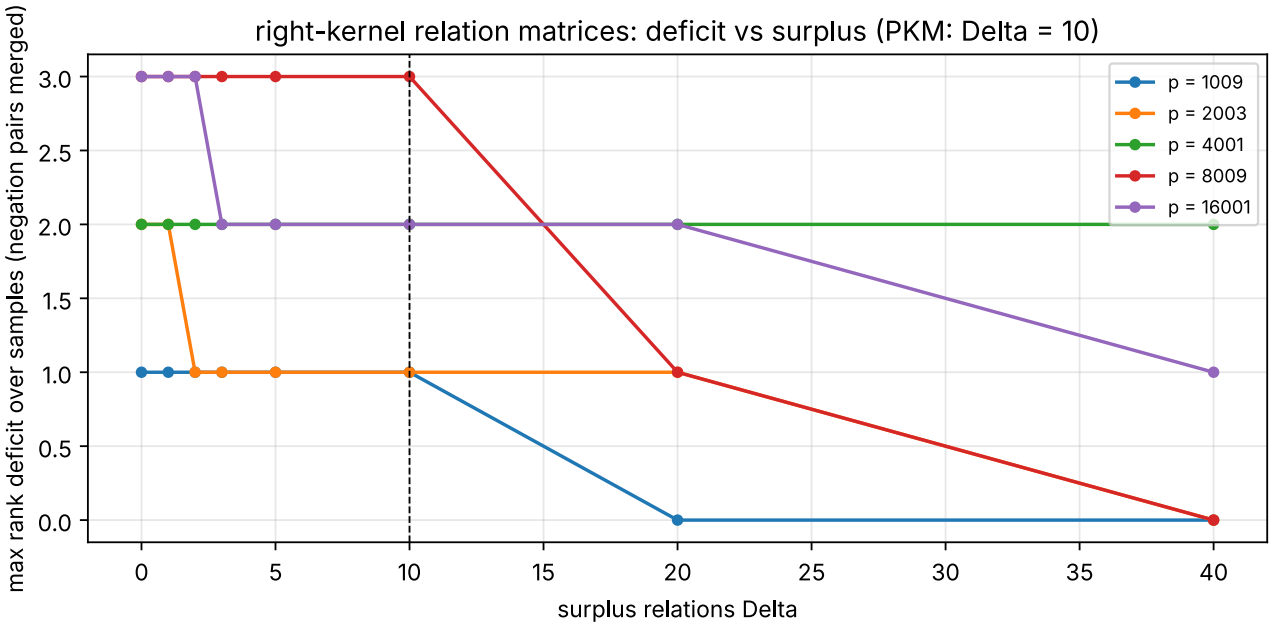


Figure 4: Maximal rank deficit over five samples against the surplus Δ, negation pairs merged.

Ten curves, p from 1009 to 16001, B ≈ √r. At Δ = 10 full rank in all samples on 2 of 10 curves; maximal deficit 1–3 at Δ = 10 and up to 2 at Δ = 40. [pcc:unverified]

trace_pullback

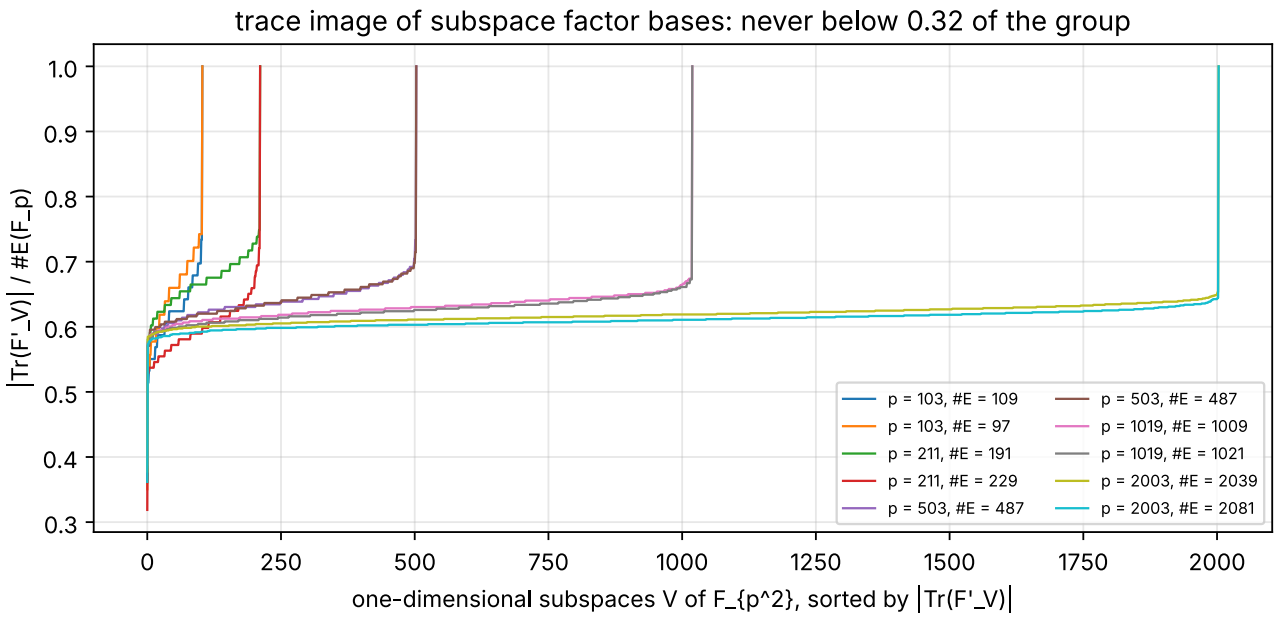


Figure 5: Trace image size of every one-dimensional subspace factor base, ten curves.

min over V of $|\mathrm{Tr}(F'_V)|\ /\#E(\mathbb{F}_p)$ is 0.32–0.39, at $V = \mathbb{F}_p \cdot i$ on every curve; median 0.60–0.67. [pcc:unverified]

ext_decomp

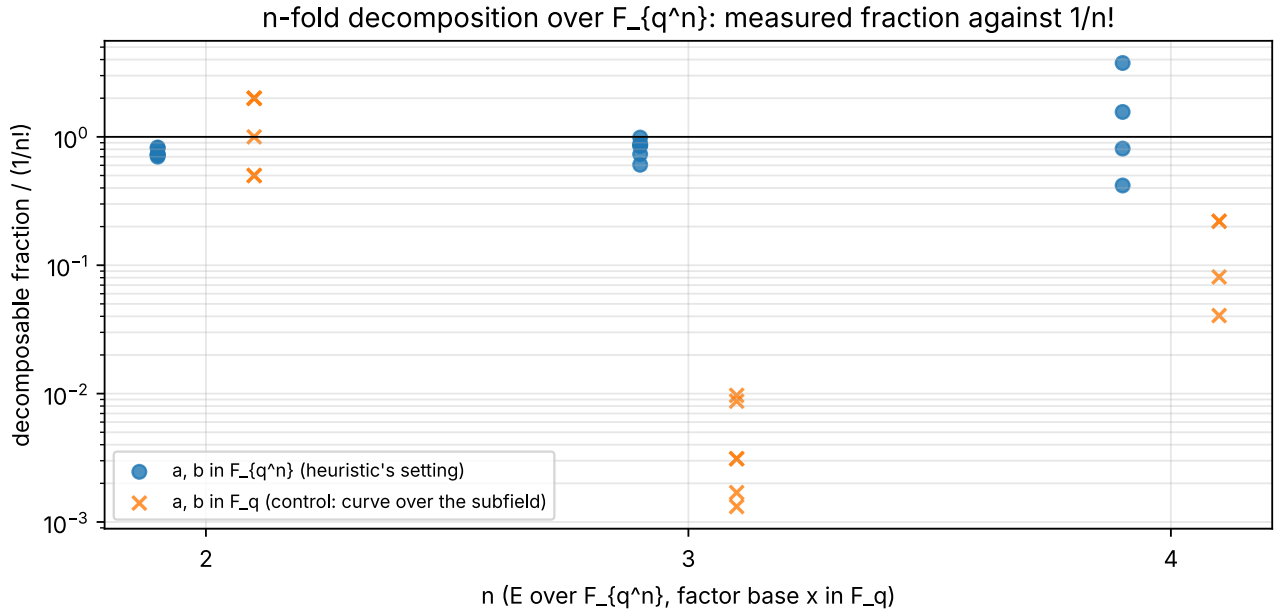


Figure 6: Decomposable fraction over $\frac{1}{n!}$; curves with $\mathbb{F}_{q^n}$ coefficients against the subfield-curve control.

0.350–0.413 at n = 2 and 0.101–0.165 at n = 3 for $\mathbb{F}_{q^n}$-coefficient curves (0.70–0.99 of $\frac{1}{n!}$); 0.0003–0.002 at n = 3 for curves defined over $\mathbb{F}_q$. [pcc:unverified]

Reproduce

`cargo run --release` in each crate directory; `python3 plots.py` for the figures; Lean: `lake env lean AxiomAudit.lean` in the Lean project.